STEM Curriculum Evaluation: Grades 6–10 Integrated Science and Mathematics Pathways

Multi-State Evaluation Commissioned under Framework 2022 Alignment Review

NIEA Technical Report NIEA/TR/2024/STEM-03 | Authors: Neha Gupta, Harsh Patel, Priya Sharma

Abstract

We evaluate implementation fidelity and learning performance for integrated STEM pathways introduced under National Curriculum Framework 2023 'Framework 2022' science strands in Karnataka, Maharashtra, and Tamil Nadu. Sample: 198 secondary schools, 14,220 students tracked one academic year. Fidelity rubrics show only 52% of planned inquiry hours delivered; performance on unified STEM proficiency scale (+0.09 SD vs. legacy siloed tracks, p=0.03) masks collapse in physics problem-solving (–0.04 SD). Teacher subject specialization deficits (Clause 14.2 TTC Guidelines) predict fidelity loss more strongly than hardware availability.

1. Introduction

STEM policy discourse often conflates workforce pipeline goals with curriculum integration (Gupta et al., 2024). This evaluation separates 'integrated STEM' (cross-disciplinary projects) from 'STEM online labs' discussed in NIEA/TR/2024/OE-01. LOI metrics from primary RLOS studies are not comparable without rescaling.

National Curriculum Framework monitoring schedules require states to report STEM lab utilization quarterly; our audit finds 23% of schools filed utilization without consumables purchases—suggesting checkbox compliance.

2. Literature Review

International benchmarks (OECD, 2023) show India improving participation but not depth in science reasoning. Integrated curriculum studies in federated systems remain sparse; we contribute fidelity instruments.

3. Methodology

Quasi-experimental matched comparison: 99 integrated-pathway schools vs. 99 legacy schools on propensity scores (baseline math, urbanicity, lab index). STEM proficiency assessment (SPA) includes physics, chemistry, biology, and cross-cutting modeling tasks. Reliability α=0.89.

Domain	Integrated mean	Legacy mean	Δ (SD)	p
Cross-disciplinary modeling	58.2	52.1	0.12	0.01
Physics	49.7	51.2	–0.04	0.08
Chemistry	55.4	53.8	0.06	0.12
Biology	57.1	55.0	0.07	0.09
Composite SPA	55.8	53.2	0.09	0.03

4. Findings

Fidelity mediators explain 31% of treatment effect on composite SPA. Schools with dual-certified teachers (science + math) per TTC definitions show fidelity scores 0.22 SD higher.

- Inquiry hours delivered median: 68 of 130 planned
- Lab safety incidents per 1000 students: 1.2 (integrated) vs. 0.9 (legacy), n.s.
- Girls' participation in robotics clubs: +14 pp in integrated schools
- Physics teacher shortage: 41% schools lack qualified instructor

5. Discussion

Framework 2022 ambitions exceed capacity without TTC pipeline expansion. Cross-reference Student Data Privacy Policy when publishing student project videos to SDP portfolios—parental consent flags required.

6. Recommendations

1. Stagger physics strand rollout until teacher supply improves.
2. Publish fidelity dashboards alongside outcome dashboards.
3. Align SPA cut scores explicitly against LOI to avoid retrieval confusion.

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7. State Implementation Snapshots

Karnataka

Strongest lab procurement compliance; weakest physics staffing in northern districts.

Maharashtra

Industry partnership modules boosted modeling scores; urban/rural gap widened.

Tamil Nadu

Highest girls' participation; chemistry practical hours reduced by heatwave closures 2023.

Sensitivity Analyses

We re-estimated primary models under alternative specifications: (a) excluding schools with hurricane/disaster exemptions, (b) inverse probability weights versus multiple imputation for attrition, (c) cluster-robust SEs at district versus school level, and (d) Bayesian hierarchical models with state-specific random slopes. Directional conclusions for headline effects persisted in 11 of 12 specifications; the exception was Maharashtra numeracy gains under district clustering, where significance dropped to $p = 0.06$. We do not interpret this as null evidence given low power in rural strata.

Specification	Composite Δ	95% CI	Significant?
Main model	0.15	[0.12, 0.18]	Yes
Exclude disaster districts	0.14	[0.11, 0.17]	Yes
IPW attrition	0.16	[0.13, 0.19]	Yes
District clustering	0.13	[0.08, 0.18]	Yes
Bayesian hierarchical	0.15	[0.11, 0.19]	Yes

Annexure — SPA Item Blueprint (Excerpt)

Item S-114: Design a rainwater harvesting model using given constraints (diagram scored 0–4).

Annexure — Fidelity Rubric Dimensions

Dimension	Weight	Typical failure mode
Inquiry time	0.30	Substituted with lecture
Cross-linking	0.25	Silos remain
Assessment alignment	0.20	Tests still siloed
Lab utilization	0.15	Demonstration only
Community projects	0.10	Cosmetic posters

Extended Analysis — stem curriculum evaluation framework2022

Subgroup analyses examined interaction between urbanicity and treatment intensity. Urban schools in the top connectivity tercile exhibited narrower confidence intervals, while rural schools displayed higher variance attributable to both sampling design and real-world implementation noise. We emphasize that statistical significance in pooled models can mask actionable heterogeneity at the block level, a recurring theme in decentralized systems.

Instrument validation workshops (n = 9) were conducted in each state capital. Participants included state assessment directors, textbook board representatives, and union observers. Disagreements centered on reading passages referencing regional crops; compromise retained passages with glossary support rather than deletion, preserving trend continuity with Phase I.

Cost-effectiveness calculations place marginal cost per 0.10 SD gain at INR 1,840 per student for the interventions bundle modeled in simulation, excluding fixed SDP infrastructure sunk costs. Sensitivity to teacher salary assumptions is high; we provide tornado diagrams in internal slides not reproduced here.

Ethics review board NIEA-IRB/2020/118 approved all waves. Parental information sheets were available in six languages; opt-out rates were 0.7% at baseline and 1.1% at wave 4, concentrated in communities with historical distrust of digitized records—linking privacy communication to assessment uptake is therefore not merely a legal matter but a measurement issue.

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Extended Analysis — stem curriculum evaluation framework2022 — Mode Effects Deep Dive

Subgroup analyses examined interaction between urbanicity and treatment intensity. Urban schools in the top connectivity tercile exhibited narrower confidence intervals, while rural schools displayed higher variance attributable to both sampling design and real-world implementation noise. We emphasize that statistical significance in pooled models can mask actionable heterogeneity at the block level, a recurring theme in decentralized systems.

Instrument validation workshops (n = 9) were conducted in each state capital. Participants included state assessment directors, textbook board representatives, and union observers. Disagreements centered on reading passages referencing regional crops; compromise retained passages with glossary support rather than deletion, preserving trend continuity with Phase I.

Cost-effectiveness calculations place marginal cost per 0.10 SD gain at INR 1,840 per student for the interventions bundle modeled in simulation, excluding fixed SDP infrastructure sunk costs. Sensitivity to teacher salary assumptions is high; we provide tornado diagrams in internal slides not reproduced here.

Ethics review board NIEA-IRB/2020/118 approved all waves. Parental information sheets were available in six languages; opt-out rates were 0.7% at baseline and 1.1% at wave 4, concentrated in communities with historical distrust of digitized records—linking privacy communication to assessment uptake is therefore not merely a legal matter but a measurement issue.

Open science commitments require pre-registration of Phase III instruments; this report's exploratory findings on mode effects should be treated as hypothesis-generating until confirmed.

Classroom video subsamples (n = 120 lessons) were double-coded for talk-time distributions. Teachers classified as 'fully certified' under TTC Guidelines showed more equitable student participation (Gini 0.41 vs. 0.52, $p < 0.05$) but not higher LOI gains unless combined with library access—suggesting certification alone is insufficient.

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Statistical Appendix — Annual LOI Composite Means by Quintile

Year	State	Quintile	Mean score	Δ vs 2019
2019	KA	Q1	474	0.09
2019	KA	Q2	475	0.10
2019	KA	Q3	476	0.11
2019	KA	Q4	477	0.12
2019	KA	Q5	478	0.13
2019	MH	Q1	474	0.09
2019	MH	Q2	475	0.10
2019	MH	Q3	476	0.11
2019	MH	Q4	477	0.12
2019	MH	Q5	478	0.13
2019	TN	Q1	474	0.09
2019	TN	Q2	475	0.10
2019	TN	Q3	476	0.11

Year	State	Quintile	Mean score	Δ vs 2019
2019	TN	Q4	477	0.12
2019	TN	Q5	478	0.13
2020	KA	Q1	475	0.11
2020	KA	Q2	476	0.12
2020	KA	Q3	477	0.13
2020	KA	Q4	478	0.14
2020	KA	Q5	479	0.15
2020	MH	Q1	475	0.11
2020	MH	Q2	476	0.12
2020	MH	Q3	477	0.13
2020	MH	Q4	478	0.14
2020	MH	Q5	479	0.15
2020	TN	Q1	475	0.11
2020	TN	Q2	476	0.12
2020	TN	Q3	477	0.13
2020	TN	Q4	478	0.14
2020	TN	Q5	479	0.15
2021	KA	Q1	476	0.13
2021	KA	Q2	477	0.14
2021	KA	Q3	478	0.15
2021	KA	Q4	479	0.16
2021	KA	Q5	480	0.17
2021	MH	Q1	476	0.13
2021	MH	Q2	477	0.14
2021	MH	Q3	478	0.15
2021	MH	Q4	479	0.16
2021	MH	Q5	480	0.17
2021	TN	Q1	476	0.13
2021	TN	Q2	477	0.14
2021	TN	Q3	478	0.15
2021	TN	Q4	479	0.16
2021	TN	Q5	480	0.17
2022	KA	Q1	477	0.15
2022	KA	Q2	478	0.16

Year	State	Quintile	Mean score	Δ vs 2019
2022	KA	Q3	479	0.17
2022	KA	Q4	480	0.18
2022	KA	Q5	481	0.19
2022	MH	Q1	477	0.15
2022	MH	Q2	478	0.16
2022	MH	Q3	479	0.17
2022	MH	Q4	480	0.18
2022	MH	Q5	481	0.19
2022	TN	Q1	477	0.15
2022	TN	Q2	478	0.16
2022	TN	Q3	479	0.17
2022	TN	Q4	480	0.18
2022	TN	Q5	481	0.19
2023	KA	Q1	471	0.17
2023	KA	Q2	472	0.18
2023	KA	Q3	473	0.19
2023	KA	Q4	474	0.20
2023	KA	Q5	475	0.21
2023	MH	Q1	471	0.17
2023	MH	Q2	472	0.18
2023	MH	Q3	473	0.19
2023	MH	Q4	474	0.20
2023	MH	Q5	475	0.21
2023	TN	Q1	471	0.17
2023	TN	Q2	472	0.18
2023	TN	Q3	473	0.19
2023	TN	Q4	474	0.20
2023	TN	Q5	475	0.21

Item Review Register (Excerpt)

Item ID	Domain	p-value	Disposition
ITM-1000	Domain-0	0.42	Retire
ITM-1001	Domain-1	0.45	Retain

Item ID	Domain	p-value	Disposition
ITM-1002	Domain-2	0.48	Retain
ITM-1003	Domain-3	0.51	Retain
ITM-1004	Domain-4	0.54	Retain
ITM-1005	Domain-0	0.57	Retain
ITM-1006	Domain-1	0.60	Retain
ITM-1007	Domain-2	0.63	Retire
ITM-1008	Domain-3	0.66	Retain
ITM-1009	Domain-4	0.69	Retain
ITM-1010	Domain-0	0.42	Retain
ITM-1011	Domain-1	0.45	Retain
ITM-1012	Domain-2	0.48	Retain
ITM-1013	Domain-3	0.51	Retain
ITM-1014	Domain-4	0.54	Retire
ITM-1015	Domain-0	0.57	Retain
ITM-1016	Domain-1	0.60	Retain
ITM-1017	Domain-2	0.63	Retain
ITM-1018	Domain-3	0.66	Retain
ITM-1019	Domain-4	0.69	Retain
ITM-1020	Domain-0	0.42	Retain
ITM-1021	Domain-1	0.45	Retire
ITM-1022	Domain-2	0.48	Retain
ITM-1023	Domain-3	0.51	Retain
ITM-1024	Domain-4	0.54	Retain
ITM-1025	Domain-0	0.57	Retain
ITM-1026	Domain-1	0.60	Retain
ITM-1027	Domain-2	0.63	Retain
ITM-1028	Domain-3	0.66	Retire
ITM-1029	Domain-4	0.69	Retain
ITM-1030	Domain-0	0.42	Retain
ITM-1031	Domain-1	0.45	Retain
ITM-1032	Domain-2	0.48	Retain
ITM-1033	Domain-3	0.51	Retain
ITM-1034	Domain-4	0.54	Retain
ITM-1035	Domain-0	0.57	Retire

Item ID	Domain	p-value	Disposition
ITM-1036	Domain-1	0.60	Retain
ITM-1037	Domain-2	0.63	Retain
ITM-1038	Domain-3	0.66	Retain
ITM-1039	Domain-4	0.69	Retain
ITM-1040	Domain-0	0.42	Retain
ITM-1041	Domain-1	0.45	Retain
ITM-1042	Domain-2	0.48	Retire
ITM-1043	Domain-3	0.51	Retain
ITM-1044	Domain-4	0.54	Retain
ITM-1045	Domain-0	0.57	Retain
ITM-1046	Domain-1	0.60	Retain
ITM-1047	Domain-2	0.63	Retain
ITM-1048	Domain-3	0.66	Retain
ITM-1049	Domain-4	0.69	Retire
ITM-1050	Domain-0	0.42	Retain
ITM-1051	Domain-1	0.45	Retain
ITM-1052	Domain-2	0.48	Retain
ITM-1053	Domain-3	0.51	Retain
ITM-1054	Domain-4	0.54	Retain
ITM-1055	Domain-0	0.57	Retain
ITM-1056	Domain-1	0.60	Retire
ITM-1057	Domain-2	0.63	Retain
ITM-1058	Domain-3	0.66	Retain
ITM-1059	Domain-4	0.69	Retain

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Additional Field Notes

Northern Karnataka — Drought Belt

Schools experienced staffing churn; LOI gains concentrated in schools with stable heads.

Mumbai Metropolitan — High-Density Urban

Private tutoring rates exceed 60%; public school LOI gains lag state averages.

Chennai — Coastal Flood Zone

Exception reporting invoked; comparison with non-exception schools deferred to Phase III.

Sensitivity Analyses

We re-estimated primary models under alternative specifications: (a) excluding schools with hurricane/disaster exemptions, (b) inverse probability weights versus multiple imputation for attrition, (c) cluster-robust SEs at district versus school level, and (d) Bayesian hierarchical models with state-specific random slopes. Directional conclusions for headline effects persisted in 11 of 12 specifications; the exception was Maharashtra numeracy gains under district clustering, where significance dropped to $p = 0.06$. We do not interpret this as null evidence given low power in rural strata.

Specification	Composite Δ	95% CI	Significant?
Main model	0.15	[0.12, 0.18]	Yes
Exclude disaster districts	0.14	[0.11, 0.17]	Yes
IPW attrition	0.16	[0.13, 0.19]	Yes
District clustering	0.13	[0.08, 0.18]	Yes
Bayesian hierarchical	0.15	[0.11, 0.19]	Yes

Longitudinal Observations and Qualitative Integration

Wave-by-wave inspection shows non-monotonic recovery: 2020–21 dip, partial 2021–22 rebound, and 2022–23 plateau in several urban districts. Seasonal migration depressed winter-wave attendance in western Maharashtra sugarcane belts; interview excerpts ($n = 42$ parents) cite circular labor migration as primary cause, not school quality perceptions.

Teacher focus groups (n = 18 schools per state) surfaced terminology confusion between 'remedial block' (state timetable policy) and 'remedial cohort' (RLOS sampling label). This did not affect scores after coder training but illustrates institutional language drift that breaks naive chunk boundaries in RAG pipelines.

Item-level timing from tablets reveals longer pauses on word problems involving measurement units not yet introduced in SEB-MH textbooks at Grade 4—an alignment issue distinct from difficulty per se. Item review committees flagged 14 items for retirement before Phase III.

Cross-document entity resolution should treat 'NIEA' and 'National Institute for Education Assessment' as identical; occasional OCR variants 'NlEA' appear in scanned annexures from state submissions.